

Temporal knowledge-gated networks for readmission and treatment-escalation risk in real-world diabetes inpatient records

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Abstract

Objective: To predict 30-day readmission and escalation of glucose-lowering treatment at the next hospital stay from real-world longitudinal inpatient records of people with diabetes, and to test whether knowledge-aware sequence models add value over strong gradient boosting under rigorous validation.

Methods: We used the public Diabetes 130-US Hospitals database (99,340 encounters of 69,987 patients after exclusions; 16,341 patients with repeat stays). We propose the Temporal Knowledge-Gated Network (TKGN), which encodes each stay with a hybrid knowledge graph (ICD-9-CM hierarchy, drug classes and a training-only co-morbidity graph), summarises earlier stays and fuses them with the current stay through history-reliability gates, and TKGN-B, which trains TKGN as a residual on an out-of-fold LightGBM logit. Ten models, including tuned logistic regression, random forest, XGBoost, LightGBM, GRU, RETAIN and Transformer baselines, were compared on five patient-disjoint splits and a prospective temporal split, with patient-clustered bootstrap confidence intervals, DeLong tests, calibration, ablation, learning-curve, explanation and subgroup analyses.

Results: Readmission and escalation prevalence were 11.4% and 35.0%. TKGN-B achieved the highest mean AUROC for readmission (0.690 versus 0.688 for LightGBM) and escalation (0.688 versus 0.687), outperforming LightGBM in 4/5 and 5/5 repeats, with good calibration (ECE 0.007 and 0.016). It was significantly better than all deep sequence baselines for readmission and the best model on the temporal escalation split (0.680 versus 0.673, $p < 0.001$); differences to gradient boosting on patient-disjoint splits were small and not statistically significant, and a random forest was most robust on the temporal readmission split. Earlier stays were the most valuable model input, whereas knowledge-graph code sharing did not improve discrimination at full data size. Test AUROC differed by at most 0.051 across adequately sized racial groups and 0.023 between sexes, and most across age bands (0.098).

Conclusion: Anchoring a knowledge-gated sequence model on gradient boosting yields calibrated risk estimates that are at least as accurate as the strongest tabular learners and clearly better than standard deep sequence models on real diabetes inpatient data. Gains over boosting are modest, underscoring the need for patient-disjoint and temporal validation when new architectures are proposed.

Keywords: electronic health records, diabetes mellitus, hospital readmission, knowledge graph, ICD-9-CM, sequence modelling, gradient boosting, calibration, algorithmic fairness

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1. Introduction

People with diabetes are admitted to hospital more often than people without it, and a notable share of those admissions is followed by an early return (Rubin, 2015). Unplanned readmission within 30 days is costly, it is used as a quality indicator, and it often reflects problems that began before discharge: an unstable glycaemic regimen, unresolved co-morbidity, or a care transition that was not prepared. A second, less studied signal is the escalation of glucose-lowering therapy between consecutive stays. Treatment intensification is the expected response to worsening glycaemic control, and its timing is known to lag behind clinical need (Khunti et al., 2013); anticipating it at discharge could help teams plan follow-up and medication review (American Diabetes Association Professional Practice Committee, 2024).

Both outcomes depend on information that is spread over time and over a large vocabulary of clinical codes. Hospital records store diagnoses as ICD-9-CM codes, and those codes are organised in a public hierarchy that groups related conditions. Many codes are rare in any single institution, so a model that learns one free parameter vector per code has little to go on for most of them. At the same time, the history available for a given patient is short and uneven: in routinely collected inpatient data, most patients appear only once, while a minority accumulate many stays. A predictive model therefore has to decide, patient by patient, how much the past should change the estimate derived from the current stay.

Most published readmission models for the widely used Diabetes 130-US Hospitals database (Strack et al., 2014) treat each encounter as an independent row, discarding the link between stays of the same person. Deep sequence models developed for large EHR corpora, such as recurrent networks (Choi et al., 2016a), reverse-time attention (Choi et al., 2016b) or Transformers (Li et al., 2020; Rasmy et al., 2021), do use this link, and ontology-aware representations such as GRAM (Choi et al., 2017) and KAME (Ma et al., 2018) address rare codes. Two practical gaps remain. First, on moderate-sized tabular clinical data, gradient-boosted decision trees are very strong and frequently match or outperform neural networks (Grinsztajn et al., 2022); new neural methods are seldom compared against well-tuned boosting under identical inputs. Second, evaluations often rely on a single random split at the encounter level, which allows the same patient to appear in training and test data and gives no estimate of variability, calibration or subgroup behaviour (Collins et al., 2015; Van Calster et al., 2019).

This paper addresses both gaps on real, publicly available hospital data. Our contributions are:

1. **A temporal knowledge-gated network (TKGN).** TKGN encodes each encounter with a hybrid clinical knowledge graph that combines the ICD-9-CM hierarchy with a co-morbidity graph estimated from training encounters, summarises the patient’s earlier stays, and fuses the past with the current stay through *history-reliability gates* that are conditioned on the amount of history and on the change since the last stay.
2. **A boosting-anchored variant (TKGN-B).** Instead of competing with gradient boosting, TKGN-B learns a residual on top of an out-of-fold LightGBM logit, so that the network only has to explain what the tree ensemble misses, chiefly the longitudinal and relational structure.
3. **Two clinically motivated tasks on the same cohort:** 30-day readmission and escalation of glucose-lowering treatment at the next stay.
4. **A leakage-controlled evaluation protocol** with patient-disjoint repeated splits, a prospective temporal split, strong tuned baselines (five tabular learners and three sequence

models), patient-clustered bootstrap confidence intervals, DeLong tests, calibration analysis, component ablations, learning curves and a subgroup fairness audit. All code and derived results are released.

2. Related work

Readmission prediction in diabetes. Systematic reviews report that most readmission models reach only moderate discrimination, with areas under the ROC curve commonly between 0.6 and 0.7 (Kansagara et al., 2011). The Diabetes 130-US Hospitals database was released together with an analysis of HbA1c measurement and readmission (Strack et al., 2014) and has since become a common benchmark. Readmission risk in diabetes is associated with prior utilisation, comorbidity burden and discharge circumstances (Rubin, 2015); these factors are recorded across stays, which motivates sequence-aware modelling.

Sequence models for EHR. Recurrent networks were among the first deep models applied to visit sequences (Choi et al., 2016a). RETAIN introduced two-level attention over reversed visit sequences to improve interpretability (Choi et al., 2016b), and Transformer encoders pre-trained on large claims or EHR corpora have since been proposed (Li et al., 2020; Rasmy et al., 2021). Large-scale studies showed that deep models trained on complete EHR can predict readmission and other outcomes (Rajkomar et al., 2018). Surveys document the breadth of this work and its reliance on large proprietary data (Shickel et al., 2018); public multitask benchmarks exist mainly for intensive-care time series (Harutyunyan et al., 2019).

Knowledge-aware code representations. GRAM represents a medical code as an attention-weighted combination of its ancestors in a clinical ontology and showed gains for rare codes (Choi et al., 2017); KAME extends this idea by using knowledge at the prediction stage (Ma et al., 2018). Graph convolution (Kipf and Welling, 2017) and co-occurrence statistics such as pointwise mutual information (Church and Hanks, 1990) are standard tools for relational structure. TKGN differs from these works in three respects: it keeps a code-specific embedding and adds ancestor information residually rather than as a convex combination; it refines category embeddings with an empirical co-morbidity graph whose propagation weights are initialised at zero; and it couples the code encoder with an explicit, history-length aware gating of longitudinal information.

Trees, networks and hybrids. Gradient-boosted trees (Friedman, 2001; Chen and Guestrin, 2016; Ke et al., 2017) remain the reference for heterogeneous tabular data (Grinsztajn et al., 2022). Hybrid designs that feed tree outputs into another learner (He et al., 2014) or combine learners by stacking (Wolpert, 1992) are well established. TKGN-B uses the boosting logit as a fixed offset and trains the network on the residual, with out-of-fold offsets on the training data to avoid optimistic anchoring. Gating as a fusion mechanism has been studied for multimodal inputs (Arevalo et al., 2017); here the gate arbitrates between the current stay and an uncertain, variable-length history.

Evaluation of clinical prediction models. Reporting guidelines ask for calibration, internal validation that respects the data structure, and transparent handling of thresholds (Collins et al., 2015). Poor calibration can make an accurate model harmful (Van Calster et al., 2019), and modern neural networks are often mis-calibrated (Guo et al., 2017). Algorithmic bias across racial groups has been documented in deployed risk tools (Obermeyer et al., 2019), which motivates subgroup audits.

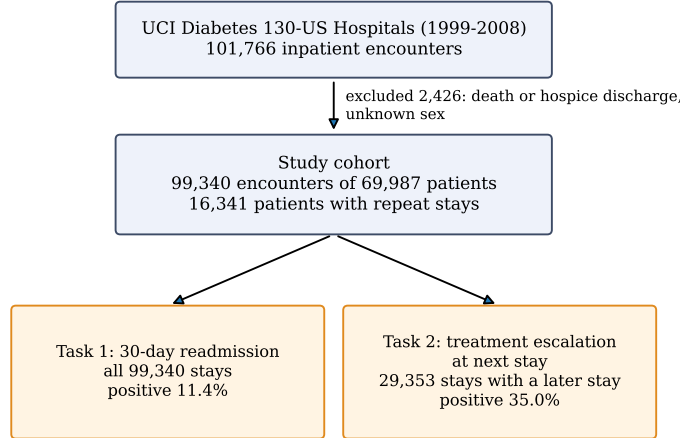


Figure 1: Cohort derivation and the two prediction tasks.

3. Data and cohort

3.1. Source

We used the Diabetes 130-US Hospitals for Years 1999–2008 database (Strack et al., 2014; Clore et al., 2014), distributed by the UCI Machine Learning Repository under a CC BY 4.0 licence. It contains 101,766 de-identified inpatient encounters of patients with diabetes from 130 US hospitals and integrated delivery networks. Each encounter lasted one to fourteen days and includes demographics, admission and discharge characteristics, utilisation counts for the preceding year, up to three ICD-9-CM diagnoses, HbA1c and serum glucose test results, the prescription status of 23 glucose-lowering drugs (not prescribed, steady, dose increased, dose decreased) and the readmission outcome. A persistent patient number links repeat encounters of the same person. The file used in this study is the unmodified public release; its SHA-256 checksum is verified by the code before any processing. No record was simulated, imputed from synthetic distributions or augmented.

3.2. Cohort definition

Following Strack et al. (2014), encounters ending in death or discharge to hospice were excluded because readmission is not possible after them; three encounters with unknown sex were also removed (Fig. 1). The resulting cohort has 99,340 encounters of 69,987 patients, of whom 16,341 have two or more encounters. Encounters of a patient were ordered by encounter identifier, which increases with admission time in this release (calendar dates are not distributed). 29,353 encounters (29.5%) have at least one earlier stay. Table 1 summarises the cohort.

3.3. Prediction tasks

Each sample is an *index encounter* e_t of a patient. The model may use e_t and the patient’s strictly earlier encounters $e_{t-L}, \dots, e_{t-1}$ ($L = 10$), but never the outcome of e_t or any later encounter.

Task 1, 30-day readmission. The label is positive if the patient was readmitted within 30 days of discharge from e_t (all 99,340 encounters; prevalence 11.4%).

Table 1: Characteristics of the study cohort (encounter level).

Characteristic	Value
Inpatient encounters	99,340
Unique patients	69,987
Patients with ≥ 2 encounters	16,341
Encounters with ≥ 1 earlier stay	29,353
Maximum encounters per patient	40
Female, %	53.8
Age <50 / $50-69$ / ≥ 70 years, %	16.0 / 39.4 / 44.6
Race: Caucasian / African American / other or unknown, %	74.7 / 18.9 / 6.4
HbA1c measured during stay, %	16.9
Insulin prescribed, %	53.3
Length of stay, mean days	4.38
Medications administered, mean	16.0
Readmitted within 30 days, %	11.4
Encounters with a later stay (escalation task)	29,353
of which escalated at next stay, %	35.0

Task 2, treatment escalation at the next stay. For encounters that are followed by another encounter of the same patient (29,353 samples), the label is positive if the next encounter shows intensification of glucose-lowering therapy: insulin started when it was not prescribed at e_t , any drug with an increased dose, or a larger number of concurrently prescribed glucose-lowering drugs than at e_t (prevalence 35.0%). Treatment intensification is a recognised marker of deteriorating glycaemic control (Khunti et al., 2013).

The readmission outcome of an *earlier* encounter is known when the index encounter takes place (the readmission, if any, occurred before or at the index admission) and is therefore a legitimate input. The outcome of the index encounter is replaced by an “unknown” token for every model.

4. Methods

Fig. 2 gives an overview. We first describe the knowledge graph, then the encounter encoder shared by all neural models, the temporal module and gates of TKGN, and finally the boosting-anchored variant TKGN-B.

4.1. Hybrid clinical knowledge graph

Let $\mathcal{V}$ be the set of concept nodes. *Ontology edges* connect each ICD-9-CM code to its four-character subcategory, its three-character category, its chapter (17 numeric chapters plus the V and E supplementary classifications) and a root node; a level is skipped when it would repeat the text of the next level. Glucose-lowering drugs are linked to their pharmacological classes (biguanide, sulfonylurea, meglitinide, thiazolidinedione, alpha-glucosidase inhibitor, insulin). A node is retained only if it is observed in at least five training encounters, so that rare or unseen codes are represented through their closest retained ancestors.

Co-morbidity edges connect three-character categories i, j that co-occur in the same training encounter more often than expected. With n_{ij} the number of training encounters listing both

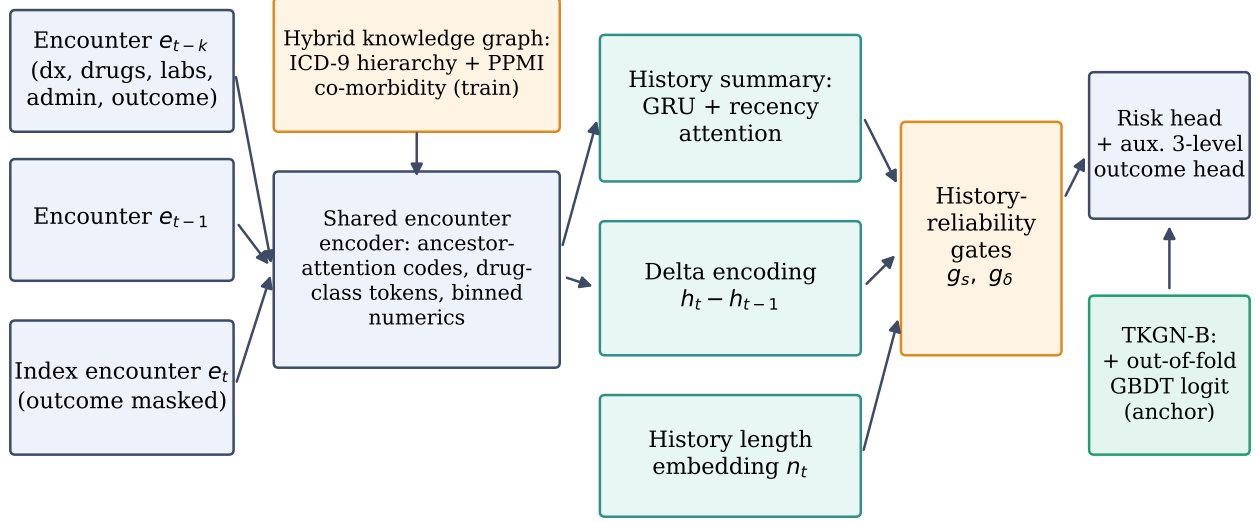


Figure 2: Overview of TKGn and TKGn-B. Every encounter is encoded with knowledge-grounded code embeddings; earlier encounters are summarised and combined with the index encounter through history-reliability gates. TKGn-B adds the network output to an out-of-fold gradient boosting logit.

categories and N the number of training encounters,

$$\text{PPMI}(i, j) = \max\left(0, \log \frac{n_{ij} N}{n_i n_j}\right), \quad (1)$$

computed only for pairs with $n_{ij} \geq 20$ and pruned to the ten strongest partners of each node. The symmetric normalised adjacency with self-loops is $\hat{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{P} + \mathbf{I})\mathbf{D}^{-1/2}$, where $\mathbf{P}$ holds the PPMI weights. Both edge types are estimated from the training partition only.

4.2. Encounter encoder

Knowledge-grounded code embeddings. Let $\mathbf{E} \in \mathbb{R}^{|\mathcal{V}| \times d}$ be free node embeddings. One propagation step over the co-morbidity graph gives

$$\tilde{\mathbf{E}} = \mathbf{E} + \tanh(\hat{\mathbf{A}} \mathbf{E} \mathbf{W}_c), \quad \mathbf{W}_c \leftarrow \mathbf{0} \text{ at initialisation}, \quad (2)$$

so the graph changes a representation only when training finds it useful. For a code c with retained own node v_c (if any) and proper ancestors $\mathcal{A}(c)$,

$$\alpha_{c,a} = \text{softmax}_{a \in \mathcal{A}(c)} \left(\mathbf{w}^\top \tanh(\mathbf{W}_\alpha [\tilde{\mathbf{e}}_{v_c}; \tilde{\mathbf{e}}_a] + \mathbf{b}_\alpha) \right), \quad \mathbf{g}_c = \tilde{\mathbf{e}}_{v_c} + \sum_{a \in \mathcal{A}(c)} \alpha_{c,a} \tilde{\mathbf{e}}_a, \quad (3)$$

with $\tilde{\mathbf{e}}_{v_c} = \mathbf{0}$ for codes without a retained node. Unlike the convex combination of GRAM (Choi et al., 2017), the residual form keeps frequent codes specific while rare codes borrow strength from the hierarchy. The three diagnosis slots of an encounter receive learned position embeddings and are pooled by attention.

Medications. Each drug k is a token $\mathbf{u}_k + \mathbf{R}^\top \mathbf{p}_k + \mathbf{s}_{\sigma_k}$, where $\mathbf{p}_k$ is the (normalised) class-membership vector, $\mathbf{R}$ holds class embeddings and $\sigma_k \in \{\text{steady, up, down}\}$ is the prescription state. Active drugs are averaged; a learned vector represents “no glucose-lowering drug”.

154 *Administrative, demographic and laboratory fields.* Twelve categorical fields (race, sex, age band,
 155 admission type, discharge disposition, admission source, payer, specialty, HbA1c result, glucose
 156 result, medication change, diabetes medication) are embedded individually ($d_f = 16$) and con-
 157 catenated. The eight counts (length of stay, laboratory procedures, procedures, medications,
 158 outpatient, emergency and inpatient visits in the prior year, number of diagnoses) are represented
 159 by quantile-bin embeddings together with their log-transformed standardised value (Gorishniy
 160 et al., 2022). All vocabularies, bin edges and scaling constants are fitted on training encounters.

161 *Outcome token.* For earlier encounters the known readmission outcome (none, after 30 days,
 162 within 30 days) is embedded; the index encounter always receives the “unknown” token. The
 163 five parts are concatenated and mapped by a two-layer perceptron with layer normalisation to
 164 $\mathbf{h} \in \mathbb{R}^d$.

165 4.3. Temporal knowledge-gated network

166 Let $\mathbf{h}_t$ be the index encounter and $\mathbf{H} = [\mathbf{h}_{t-n_t}, \dots, \mathbf{h}_{t-1}]$ the $n_t \leq L$ earlier encounters.
 167 TKGn computes three views of the past.

168 *History summary.* A GRU (Cho et al., 2014) run over $\mathbf{H}$ yields its last state $\mathbf{s}_{\text{gru}}$. Recency-
 169 aware attention uses the index encounter as query:

$$\beta_j = \text{softmax}_j \left(\frac{(\mathbf{W}_q \mathbf{h}_t)^\top \mathbf{W}_k (\mathbf{h}_j + \mathbf{r}_{t-j})}{\sqrt{d}} \right), \quad \mathbf{s}_{\text{att}} = \sum_j \beta_j \mathbf{h}_j, \quad (4)$$

170 where $\mathbf{r}_{t-j}$ embeds how many stays ago encounter j occurred. The summary is $\mathbf{s} = \mathbf{W}_s [\mathbf{s}_{\text{gru}}; \mathbf{s}_{\text{att}}]$,
 171 or a learned vector $\mathbf{s}_\emptyset$ when $n_t = 0$.

172 *Change since the last stay.* $\boldsymbol{\delta} = \mathbf{h}_t - \mathbf{h}_{t-1}$ ($\mathbf{0}$ if $n_t = 0$) captures escalation of treatment,
 173 utilisation or diagnostic burden between consecutive stays.

174 *History-reliability gates.* With $\boldsymbol{\ell}(n_t)$ an embedding of the history length and $\mathbf{c} = [\mathbf{h}_t; \mathbf{s}; \boldsymbol{\delta}; \boldsymbol{\ell}(n_t)]$,

$$\mathbf{g}_s = \sigma(\mathbf{W}_{gs} \mathbf{c} + \mathbf{b}_{gs}), \quad \mathbf{g}_\delta = \sigma(\mathbf{W}_{g\delta} \mathbf{c} + \mathbf{b}_{g\delta}), \quad \mathbf{z} = \mathbf{h}_t + \mathbf{g}_s \odot \mathbf{s} + \mathbf{g}_\delta \odot \mathbf{W}_\delta \boldsymbol{\delta}. \quad (5)$$

175 The gates are feature-wise and depend on the patient: a single earlier stay with an unremark-
 176 able course can be down-weighted, while a long history of early readmissions can dominate the
 177 representation.

178 *Outputs and loss.* A two-layer head maps $\mathbf{z}$ to the risk logit $f(\mathbf{z})$. An auxiliary linear head
 179 predicts the three-level readmission outcome of the index stay (none, > 30 days, < 30 days),
 180 which provides a richer training signal than the binary label (Caruana, 1997). The training
 181 objective is

$$\mathcal{L} = \text{BCE}(y, \sigma(f(\mathbf{z}))) + \lambda \text{CE}(o, \text{softmax}(\mathbf{W}_o \mathbf{z})), \quad \lambda = 0.5. \quad (6)$$

182 The auxiliary head is used only during training.

183 4.4. Boosting-anchored TKGn (TKGn-B)

184 Gradient boosting captures threshold effects in counts and categorical interactions very effi-
 185 ciently, whereas TKGn brings longitudinal and relational structure. TKGn-B combines them
 186 additively on the logit scale:

$$\text{logit } P(y = 1 \mid e_{\leq t}) = \phi(\mathbf{x}_t) + f(\mathbf{z}_t), \quad (7)$$

where ϕ is a LightGBM model (Ke et al., 2017) trained on the tabular view $\mathbf{x}_t$ of the same information (Section 5.2) and f is the TKG head. ϕ is kept fixed while TKG is trained. Using in-sample boosting logits as offsets would give the network a target that is already over-fitted, so the training offsets are *out-of-fold*: training patients are split into five groups, and the offset of each training encounter comes from a LightGBM model fit on the other four groups with the hyper-parameters and number of trees selected on the validation set. Validation and test offsets come from the LightGBM model fit on all training patients. The network therefore learns a residual correction, conceptually one additional boosting stage (Friedman, 2001) implemented by a sequence model.

5. Experimental design

5.1. Validation designs

Patient-grouped split. Patients, not encounters, were randomly assigned to training (70%), validation (10%) and test (20%) partitions, stratified by whether a patient has any positive sample, so that no person contributes to more than one partition. The procedure was repeated five times with different partitions and model seeds. For the first repeat the partition sizes are 69,687 / 9,859 / 19,794 encounters (readmission) and 20,726 / 2,858 / 5,769 (escalation).

Temporal (prospective) split. Encounters were ordered by identifier and cut at the 70th and 80th percentiles: models were fitted on the earliest 70%, tuned on the next 10% and tested on the most recent 20% of admissions. A test encounter may belong to a patient seen earlier, as in deployment, but its history only contains earlier stays. For the escalation task an index encounter is kept only if its next encounter falls in the same period, so no label is derived from a later period. Three seeds were run.

5.2. Baselines

Tabular learners use a design matrix with the eight counts, one-hot categorical fields, drug states, multi-hot three-character diagnosis categories (at least 50 training occurrences), chapter indicators and hand-crafted history aggregates computed from strictly earlier stays (number of earlier stays, earlier readmissions within and after 30 days, the outcome of the last stay, mean length of stay, medications, diagnoses and laboratory procedures over earlier stays, earlier insulin use, and changes since the last stay); 524 features in total. We trained L2-regularised logistic regression, random forest (Breiman, 2001), XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017) and a multilayer perceptron. *Sequence models* use the same encounter encoder as TKG with flat code embeddings: a GRU over the encounter sequence in the spirit of Doctor AI (Choi et al., 2016a), RETAIN (Choi et al., 2016b) and a two-layer Transformer encoder (Vaswani et al., 2017). All sequence baselines receive the outcomes of earlier stays, exactly like TKG.

5.3. Training and model selection

Hyper-parameters of the tabular learners were selected on the validation partition from small grids (regularisation strength; minimum leaf size; tree depth and minimum child weight; number of leaves and minimum child samples; hidden sizes and weight decay). Boosting used a learning rate of 0.03 with early stopping on validation AUROC (up to 2,000 trees). All neural models used $d = 64$, dropout 0.3, AdamW (Loshchilov and Hutter, 2019) with learning rate 5×10^{-4} , weight decay 10^{-3} , batch size 512, gradient clipping at 1.0, and early stopping on validation AUROC (patience 4, at most 30 epochs). Architectural choices for the neural models (quantile-bin and

per-field embeddings in the shared encoder, the additive code encoder, the auxiliary head of TKG_N, regularisation) were made during development on a separate random patient partition of the same cohort, mainly using its validation data, and were frozen before the evaluation runs reported here. Because all partitions are drawn from one cohort, some patients used for development also appear in evaluation test sets; this may bias the neural models slightly upwards and is discussed as a limitation. Within the evaluation runs, the test partitions were used only for the final assessment. Models were implemented in PyTorch (Paszke et al., 2019) and scikit-learn (Pedregosa et al., 2011).

5.4. Metrics and statistical analysis

Discrimination was measured by the area under the ROC curve (AUROC, primary) and the area under the precision–recall curve (AUPRC). Calibration was measured by the Brier score, the expected calibration error over ten equal-width bins (ECE, Naeini et al., 2015) and the calibration slope and intercept from logistic recalibration (Van Calster et al., 2019). Sensitivity, specificity, positive predictive value and F1 were computed at a threshold that maximised F1 on the validation partition and was then frozen. For the grouped design we report mean and standard deviation over the five repeats. For the first test run of each design we computed 95% confidence intervals with a patient-clustered percentile bootstrap (1,000 resamples; patients rather than encounters are resampled, Efron and Tibshirani, 1993), paired bootstrap intervals for AUROC differences, and the DeLong test for correlated AUROCs (DeLong et al., 1988) using the fast algorithm of Sun and Xu (2014). The DeLong test treats encounters as independent and is therefore reported together with the clustered bootstrap. Across the five grouped repeats we also report how often a model won and a paired *t*-test on per-repeat AUROC differences.

5.5. Ablation, data efficiency, explanation and fairness

Seven TKG_N variants, each removing one component, were trained on the first 3 grouped repeats. To test the pre-specified hypothesis that knowledge-graph code sharing matters most when data are scarce, TKG_N and TKG_N without the knowledge graph were additionally retrained on 5, 10 and 25% of the training patients in the first three grouped repeats (paired subsets and seeds). Learning curves retrained LightGBM, GRU, TKG_N and TKG_N-B on 5, 10, 25 and 50% of the training patients of the first repeat with the same test patients. Global explanations were obtained by permutation importance of input groups for TKG_N (AUROC drop over three permutations) and exact TreeSHAP values (Lundberg et al., 2020) for the LightGBM anchor of TKG_N-B. Subgroup performance was examined by race, sex, age band, number of earlier stays and HbA1c testing (Obermeyer et al., 2019).

6. Results

6.1. Discrimination and calibration

Tables 2 and 3 and Fig. 3 report test performance. For 30-day readmission under the patient-grouped design, TKG_N-B reached a mean AUROC of 0.690 (± 0.005) and AUPRC of 0.250, the highest of all models, followed closely by XGBoost (0.688) and LightGBM (0.688). The three sequence baselines were tightly grouped (GRU 0.682, RETAIN 0.683, Transformer 0.683), and TKG_N on its own matched them (0.682) without surpassing the boosting models. Logistic regression (0.669) and the multilayer perceptron (0.653) were clearly weaker. For treatment escalation the ranking was similar: TKG_N-B 0.688 (± 0.013), LightGBM 0.687, XGBoost 0.687,

Table 2: Test performance for 30-day readmission. Grouped split: mean $\pm$ SD over five patient-disjoint repeats; temporal split: mean over three seeds. Slope: calibration slope; Sens./Spec.: at the validation-selected threshold. Best AUROC per design in bold.

Model	AUROC	Patient-grouped split (5 repeats, mean $\pm$ SD)						Temporal split	
		AUPRC	Brier	ECE	Slope	Sens.	Spec.	AUROC	AUPRC
Logistic regression	0.669 $\pm$ 0.005	0.223 $\pm$ 0.007	0.096 $\pm$ 0.001	0.006 $\pm$ 0.001	0.85	0.41	0.81	0.603	0.190
Random forest	0.682 $\pm$ 0.005	0.233 $\pm$ 0.011	0.096 $\pm$ 0.001	0.008 $\pm$ 0.001	1.20	0.39	0.84	0.700	0.238
XGBoost	0.688 $\pm$ 0.004	0.247 $\pm$ 0.006	0.095 $\pm$ 0.001	0.004 $\pm$ 0.002	1.04	0.41	0.83	0.684	0.234
LightGBM	0.688 $\pm$ 0.004	0.247 $\pm$ 0.007	0.095 $\pm$ 0.001	0.004 $\pm$ 0.002	1.04	0.38	0.85	0.685	0.234
MLP	0.653 $\pm$ 0.009	0.212 $\pm$ 0.009	0.097 $\pm$ 0.001	0.017 $\pm$ 0.006	0.69	0.42	0.78	0.601	0.187
GRU	0.682 $\pm$ 0.005	0.236 $\pm$ 0.010	0.095 $\pm$ 0.001	0.011 $\pm$ 0.002	0.99	0.41	0.82	0.684	0.228
RETAIN	0.683 $\pm$ 0.006	0.237 $\pm$ 0.010	0.096 $\pm$ 0.001	0.015 $\pm$ 0.004	0.95	0.44	0.80	0.687	0.229
Transformer	0.683 $\pm$ 0.006	0.237 $\pm$ 0.013	0.096 $\pm$ 0.001	0.015 $\pm$ 0.008	0.86	0.41	0.82	0.690	0.231
TKGN (proposed)	0.682 $\pm$ 0.004	0.238 $\pm$ 0.011	0.095 $\pm$ 0.001	0.014 $\pm$ 0.007	0.90	0.41	0.82	0.689	0.236
TKGN-B (proposed)	0.690$\pm$0.005	0.250 $\pm$ 0.009	0.095 $\pm$ 0.001	0.007 $\pm$ 0.004	0.89	0.41	0.83	0.685	0.235

Table 3: Test performance for treatment escalation at the next stay (same layout as Table 2).

Model	AUROC	Patient-grouped split (5 repeats, mean $\pm$ SD)						Temporal split	
		AUPRC	Brier	ECE	Slope	Sens.	Spec.	AUROC	AUPRC
Logistic regression	0.666 $\pm$ 0.013	0.506 $\pm$ 0.007	0.211 $\pm$ 0.003	0.018 $\pm$ 0.004	0.85	0.82	0.37	0.646	0.497
Random forest	0.685 $\pm$ 0.012	0.531 $\pm$ 0.010	0.208 $\pm$ 0.003	0.034 $\pm$ 0.006	1.46	0.78	0.47	0.671	0.514
XGBoost	0.687 $\pm$ 0.013	0.535 $\pm$ 0.010	0.205 $\pm$ 0.003	0.015 $\pm$ 0.005	1.07	0.81	0.42	0.673	0.519
LightGBM	0.687 $\pm$ 0.012	0.536 $\pm$ 0.012	0.205 $\pm$ 0.003	0.014 $\pm$ 0.008	1.07	0.80	0.45	0.673	0.516
MLP	0.646 $\pm$ 0.007	0.485 $\pm$ 0.004	0.217 $\pm$ 0.002	0.046 $\pm$ 0.010	0.62	0.77	0.42	0.622	0.467
GRU	0.681 $\pm$ 0.014	0.527 $\pm$ 0.014	0.208 $\pm$ 0.004	0.032 $\pm$ 0.011	0.86	0.79	0.44	0.670	0.516
RETAIN	0.677 $\pm$ 0.013	0.517 $\pm$ 0.015	0.209 $\pm$ 0.004	0.034 $\pm$ 0.011	0.85	0.75	0.50	0.668	0.518
Transformer	0.679 $\pm$ 0.013	0.526 $\pm$ 0.010	0.209 $\pm$ 0.005	0.040 $\pm$ 0.015	0.76	0.78	0.46	0.668	0.516
TKGN (proposed)	0.678 $\pm$ 0.012	0.525 $\pm$ 0.009	0.209 $\pm$ 0.005	0.035 $\pm$ 0.017	0.82	0.77	0.47	0.668	0.514
TKGN-B (proposed)	0.688$\pm$0.013	0.537 $\pm$ 0.011	0.205 $\pm$ 0.004	0.016 $\pm$ 0.006	0.99	0.76	0.49	0.680	0.525

random forest 0.685, sequence models between 0.677 and 0.681, and TKGN 0.678. The absolute level of discrimination is modest for both tasks, in line with the range usually reported for readmission (Kansagara et al., 2011).

Calibration was good for the boosting models and for TKGN-B (Fig. 4). On readmission, TKGN-B had an ECE of 0.007 and a calibration slope of 0.89, compared with 0.004 and 1.04 for LightGBM; the neural sequence models were somewhat less well calibrated (ECE between 0.011 and 0.015). A slope slightly below one for TKGN-B indicates mildly over-dispersed risks, a pattern that could be removed by post-hoc recalibration on validation data. At the validation-selected thresholds, TKGN-B detected 0.41 of readmissions at a specificity of 0.83.

6.2. Statistical comparison

Table 4 compares TKGN-B with every other model. Against the deep sequence models, the advantage of TKGN-B was consistent and statistically significant for readmission: it won all five repeats against GRU, RETAIN, the Transformer and TKGN, and on the first test set the AUROC difference to TKGN was 0.0111 (95% CI 0.0060 to 0.0161; DeLong $p < 0.001$). For escalation it also won all five repeats against every sequence model (mean difference to TKGN +0.0101, paired t -test p 0.003), although the single-split bootstrap intervals, which are wider for this smaller task, included zero.

Against the tuned gradient-boosting models the differences were small. For readmission TKGN-B exceeded LightGBM in 4/5 repeats (mean Δ AUROC +0.0020, p 0.123) and XG-

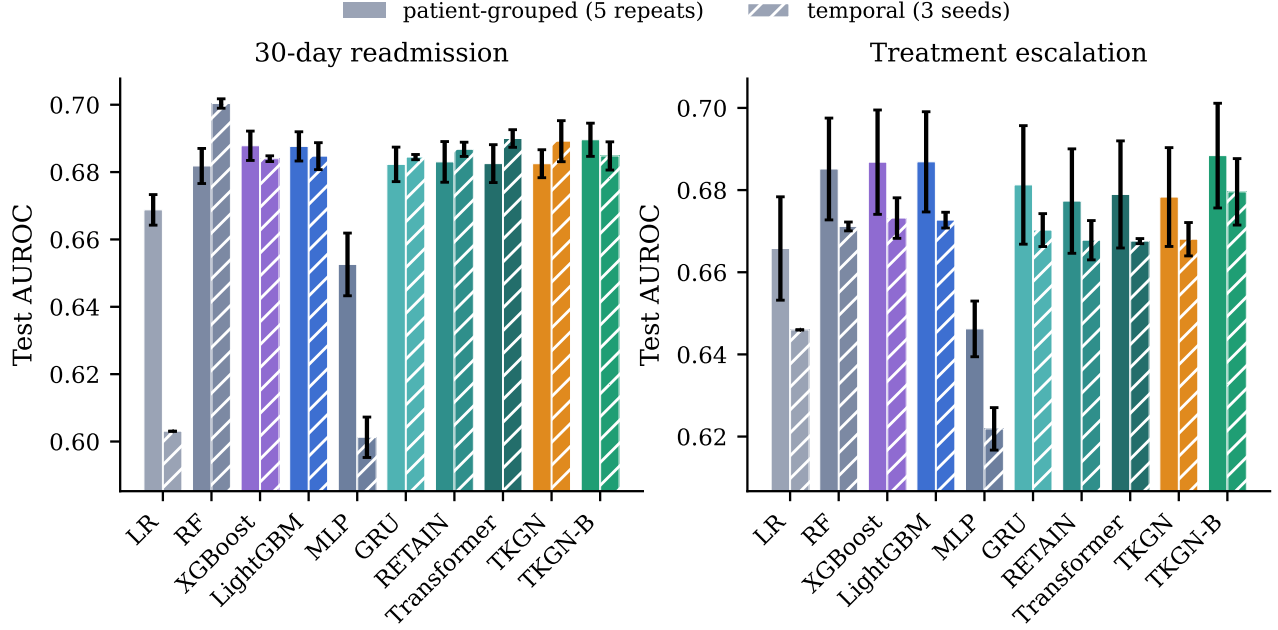


Figure 3: Test AUROC of all models. Solid bars: patient-grouped split (mean $\pm$ SD over five repeats); hatched bars: temporal split (mean $\pm$ SD over three seeds).

Boost in 4/5 (mean +0.0018); for escalation it exceeded both in 5/5 repeats (mean +0.0015 and +0.0016; p 0.088 and 0.093). None of these differences reached significance on the first test set (bootstrap p 0.916 and 0.364 versus LightGBM). We therefore interpret TKG-B as matching the strongest tabular learners with a small but consistent advantage, not as a decisive improvement over them.

6.3. Temporal validation and dataset shift

The temporal split tests models on the most recent admissions after fitting them on older ones. For treatment escalation TKG-B was again the best model (AUROC 0.680 versus 0.673 for XGBoost and 0.673 for LightGBM), and on the first seed its advantage was significant against every comparator (e.g. Δ AUROC 0.0103 versus LightGBM, DeLong p <0.001). For readmission the picture differed: random forest achieved the highest AUROC (0.700), while TKG-B (0.685), LightGBM (0.685) and XGBoost (0.684) were close to each other and TKG alone reached 0.689.

The temporal test period differs markedly from the training period. The maximum number of recorded diagnoses per stay rose from 9 to 16 (mean 7.1 versus 8.1), the share of stays preceded by an earlier stay rose from 26.1% to 37.2%, the longest history grew from 22 to 39 earlier stays, and emergency visits in the prior year nearly doubled (0.16 versus 0.30). Models that extrapolate linearly suffered most: logistic regression fell to an AUROC of 0.603 with a calibration slope of 0.07, and the MLP to 0.601. Tree ensembles, whose predictions are bounded outside the training range, and sequence models, which bound history through a fixed window and learned summaries, were much more robust.

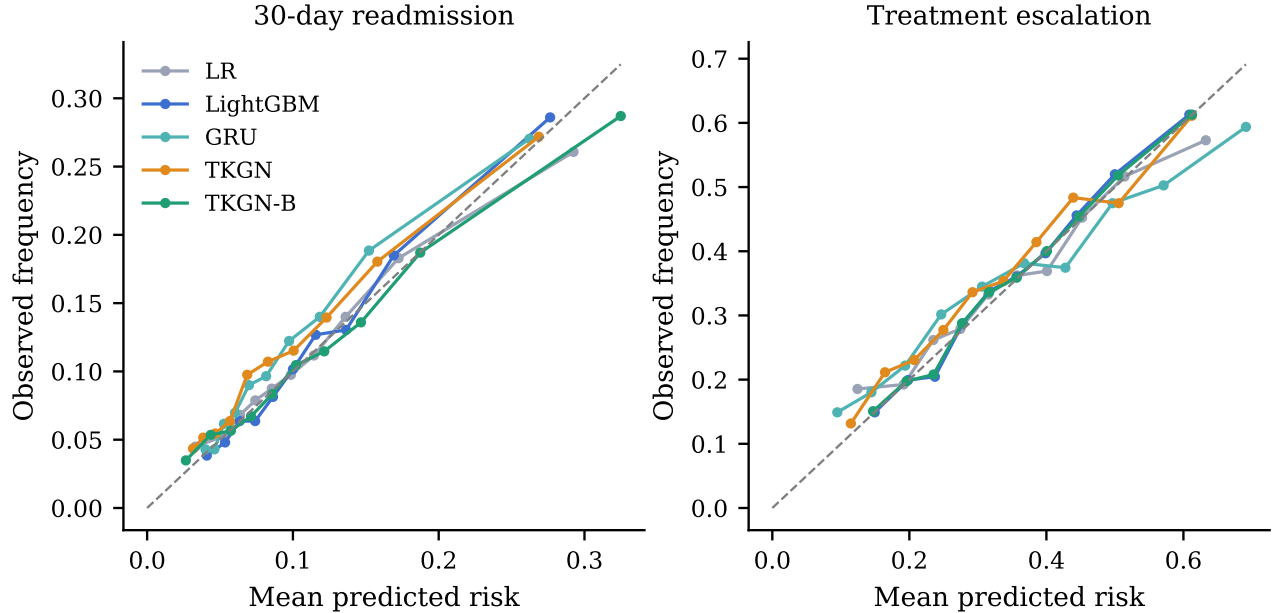


Figure 4: Reliability curves on the first grouped test set (ten equal-frequency bins).

Table 4: TKGn-B compared with each model. ΔAUROC and its 95% patient-clustered bootstrap CI refer to the first grouped test set; p_{DeLong} : DeLong test on the same set; Wins: repeats (of five) in which TKGn-B had the higher test AUROC; p_{temp} : DeLong test on the temporal test set (first seed). Positive values favour TKGn-B.

Comparator	30-day readmission					Treatment escalation				
	ΔAUROC [95% CI]	p_{DeLong}	Wins	p_{temp}		ΔAUROC [95% CI]	p_{DeLong}	Wins	p_{temp}	
Logistic regression	+0.0224 [+0.0152, +0.0299]	<0.001	5/5	<0.001		+0.0240 [+0.0145, +0.0330]	<0.001	5/5	<0.001	
Random forest	+0.0058 [-0.0004, +0.0114]	0.055	5/5	<0.001		+0.0029 [-0.0020, +0.0076]	0.237	5/5	0.003	
XGBoost	+0.0017 [-0.0028, +0.0063]	0.472	4/5	0.001		+0.0009 [-0.0025, +0.0045]	0.593	5/5	<0.001	
LightGBM	+0.0002 [-0.0037, +0.0038]	0.901	4/5	0.159		+0.0003 [-0.0004, +0.0011]	0.359	5/5	<0.001	
MLP	+0.0386 [+0.0297, +0.0472]	<0.001	5/5	<0.001		+0.0339 [+0.0227, +0.0450]	<0.001	5/5	<0.001	
GRU	+0.0061 [+0.0004, +0.0115]	0.028	5/5	0.087		+0.0072 [-0.0010, +0.0156]	0.120	5/5	0.001	
RETAIN	+0.0076 [+0.0017, +0.0132]	0.006	5/5	0.405		+0.0043 [-0.0048, +0.0134]	0.367	5/5	<0.001	
Transformer	+0.0104 [+0.0047, +0.0160]	<0.001	5/5	0.796		+0.0026 [-0.0056, +0.0112]	0.558	5/5	0.001	
TKGn (proposed)	+0.0111 [+0.0060, +0.0161]	<0.001	5/5	0.562		+0.0045 [-0.0047, +0.0136]	0.332	5/5	0.011	

6.4. Ablation

Table 5 and Fig. 5 show the effect of removing each TKGn component (first 3 grouped repeats). Access to earlier encounters was by far the most important element: without it, AUROC dropped by -0.0073 for readmission and -0.0179 for escalation. Removing the outcomes of earlier stays or the auxiliary outcome head reduced readmission AUROC by -0.0021 and -0.0021, but not escalation AUROC. The history-reliability gates and the delta encoding had effects within ± 0.001 , i.e. within run-to-run variation. Contrary to our expectation, the knowledge-graph components did not help with the full training set: flat code embeddings changed AUROC by +0.0011 (readmission) and +0.0038 (escalation), and removing only the co-morbidity propagation gave +0.0042 and +0.0063, i.e. the hierarchy-only variant was the best TKGn configuration on test data. Because this was observed after the design had been fixed on validation data, we report the pre-specified configuration as TKGn and treat the hierarchy-only variant as a finding to be confirmed in future work.

Table 5: Ablation of TKGn (mean $\pm$ SD test AUROC over the first 3 grouped repeats; Δ relative to the full model).

Variant	30-day readmission		Treatment escalation	
	AUROC	Δ	AUROC	Δ
Full TKGn	0.6847 $\pm$ 0.0035	–	0.6781 $\pm$ 0.0151	–
– knowledge graph (flat code embeddings)	0.6857 $\pm$ 0.0038	+0.0011	0.6819 $\pm$ 0.0172	+0.0038
– co-morbidity (PPMI) propagation	0.6888 $\pm$ 0.0036	+0.0042	0.6843 $\pm$ 0.0162	+0.0063
– earlier encounters (index stay only)	0.6773 $\pm$ 0.0059	-0.0073	0.6601 $\pm$ 0.0191	-0.0179
– history-reliability gates	0.6837 $\pm$ 0.0036	-0.0009	0.6778 $\pm$ 0.0158	-0.0002
– delta encoding	0.6851 $\pm$ 0.0038	+0.0005	0.6770 $\pm$ 0.0137	-0.0010
– outcomes of earlier stays	0.6826 $\pm$ 0.0033	-0.0021	0.6803 $\pm$ 0.0156	+0.0023
– auxiliary outcome head	0.6826 $\pm$ 0.0025	-0.0021	0.6819 $\pm$ 0.0202	+0.0039

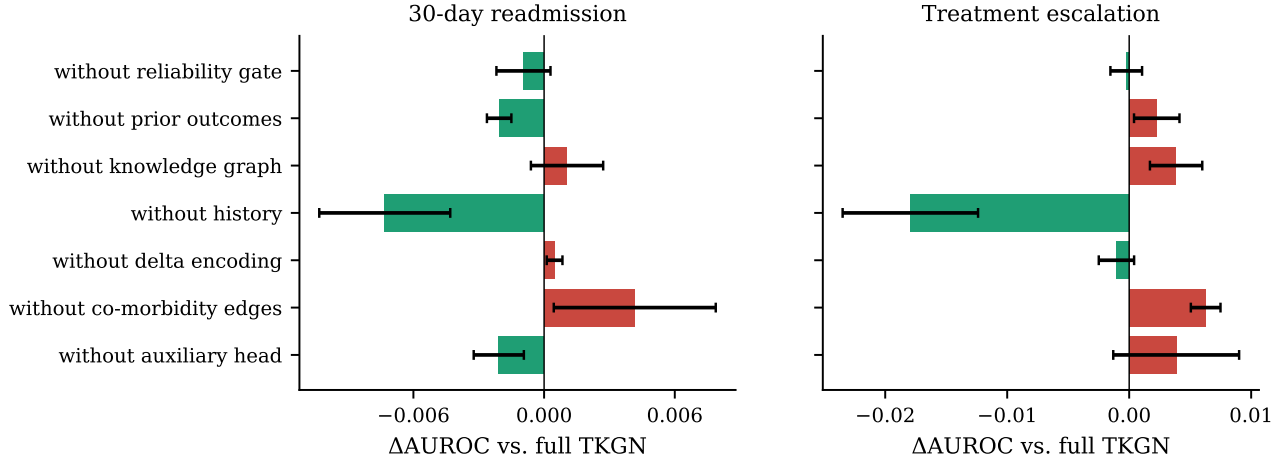


Figure 5: Change in test AUROC when one component of TKGn is removed (mean $\pm$ SD of paired differences over repeats). Green: the component helps; red: the model is better without it.

6.5. Knowledge graph and data scarcity

PENDING

6.6. Data efficiency

Fig. 6 and Table 6 show performance when only part of the training patients is available. TKGn-B was at least as good as LightGBM at every training size for readmission, with the largest gains at small sizes (5%: 0.652 versus 0.644; 25%: 0.669 versus 0.660). For escalation, LightGBM was better with 5% of the training patients (0.632 versus 0.627) and TKGn-B was better from 10% onwards (e.g. 50%: 0.674 versus 0.669). These curves come from one repeat each and should be read as trends.

6.7. Patients with and without earlier stays

For readmission, discrimination was lower for patients with a longer history, whose baseline risk is higher and more homogeneous (Table 7); for escalation it was higher (0.667 for first stays versus 0.693 with two or more earlier stays). For readmission among patients with two or more earlier stays, TKGn-B reached an AUROC of 0.620, compared with 0.608 for LightGBM, and TKGn alone reached 0.619; the sequence information therefore contributes most where the tabular history summaries saturate. For first stays the two approaches were equivalent (0.673 versus 0.671).

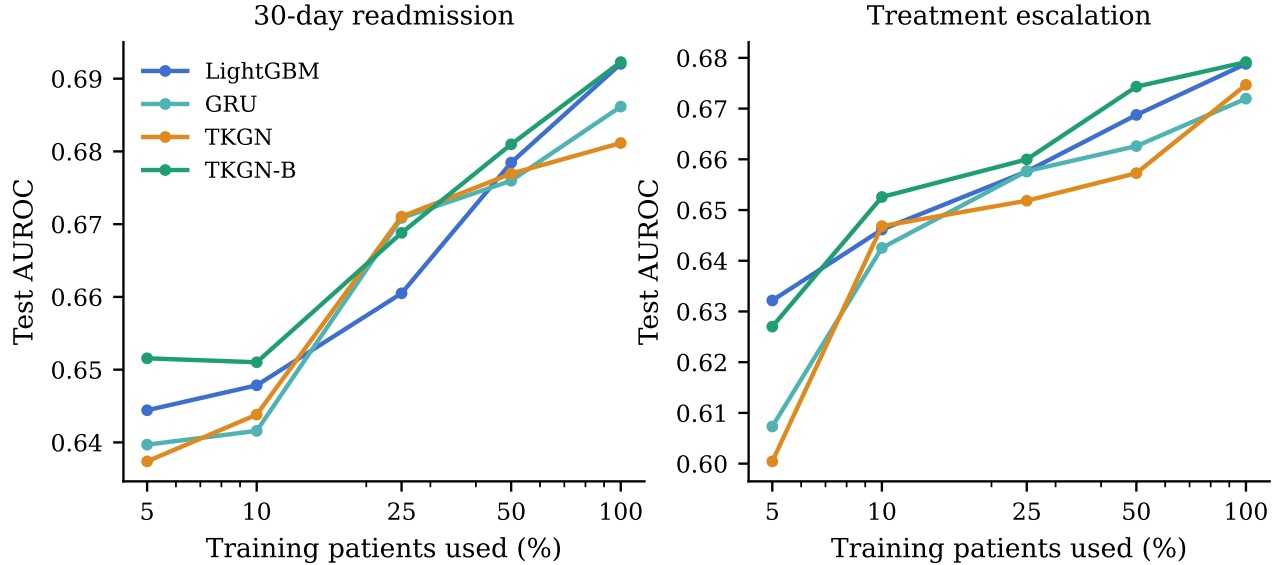


Figure 6: Test AUROC as a function of the share of training patients (first grouped repeat; same test patients at every size).

Table 6: Learning curve: test AUROC by share of training patients.

Task	Model	5%	10%	25%	50%	100%
30-day readmission	LightGBM	0.644	0.648	0.660	0.678	0.692
	GRU	0.640	0.642	0.671	0.676	0.686
	TKGn	0.637	0.644	0.671	0.677	0.681
	TKGn-B	0.652	0.651	0.669	0.681	0.692
Treatment escalation	LightGBM	0.632	0.646	0.658	0.669	0.679
	GRU	0.607	0.643	0.658	0.663	0.672
	TKGn	0.600	0.647	0.652	0.657	0.675
	TKGn-B	0.627	0.653	0.660	0.674	0.679

The learned gates behaved differently for the two tasks (Fig. 7). For escalation, the mean history gate increased with the number of earlier stays (0.529 with none, 0.603 with five or more), i.e. the model relied more on longer histories. For readmission, the gate was lower for patients with history than the value it applied to the learned “no history” vector (0.382 with one earlier stay, 0.486 with none), indicating that the network damped the history summary and relied on the index stay together with the outcome tokens of earlier stays.

6.8. Explanations

Permutation importance (Fig. 8) identified Utilisation counts as the most important input group of TKGn for readmission (AUROC drop 0.053), followed by the sequence of earlier encounters and the admission and discharge context; for escalation the anti-diabetic medication profile dominated (drop 0.106), again followed by earlier encounters. HbA1c and glucose results contributed almost nothing, consistent with the fact that they were measured in fewer than one in five stays. For the LightGBM anchor, TreeSHAP ranked the number of inpatient stays in the preceding year and discharge home as the strongest readmission predictors, and a steady insulin dose and the change in the number of prescribed diabetes drugs since the last stay as the strongest escalation predictors. These patterns agree with clinical knowledge on readmission

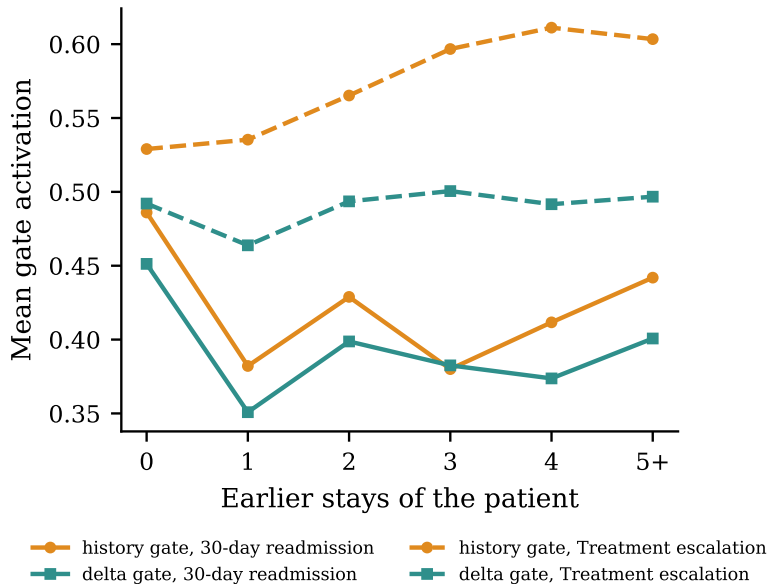


Figure 7: Mean activation of the history and delta gates by number of earlier stays (first grouped test set).

after diabetes admissions (Rubin, 2015) and support the plausibility of the models.

6.9. Subgroups and fairness

Table 7 and Fig. 9 report subgroup performance. For readmission, the AUROC of TKGN-B differed by at most 0.051 between racial groups with at least 300 test encounters (African American 0.715, Caucasian 0.684), by 0.023 between women and men, and by 0.098 between age bands, with the lowest discrimination in patients aged 70 years or older. For escalation the corresponding gaps were 0.026, 0.000 and 0.023. Calibration within subgroups was acceptable (ECE at most 0.031 in all groups with at least 300 encounters). Groups with fewer than 300 test encounters (Asian, $n = 105$) give unstable estimates and are shown for completeness only.

7. Discussion

7.1. Principal findings

On more than 99,000 real inpatient encounters, a boosting-anchored knowledge-gated sequence model (TKGN-B) was the best or joint-best model for both 30-day readmission and next-stay treatment escalation under patient-disjoint validation, was clearly and significantly better than three established deep sequence architectures, and was the best model on the prospective temporal split for treatment escalation. Its advantage over carefully tuned gradient boosting was, however, small (about 0.002 AUROC on average) and not statistically significant on a single test set, and on the temporal readmission split a random forest was more robust than every other model. The most important ingredient of the sequence model was access to the patient’s earlier encounters (and, for readmission, their outcomes); the knowledge-graph components did not improve discrimination when the full training set was used.

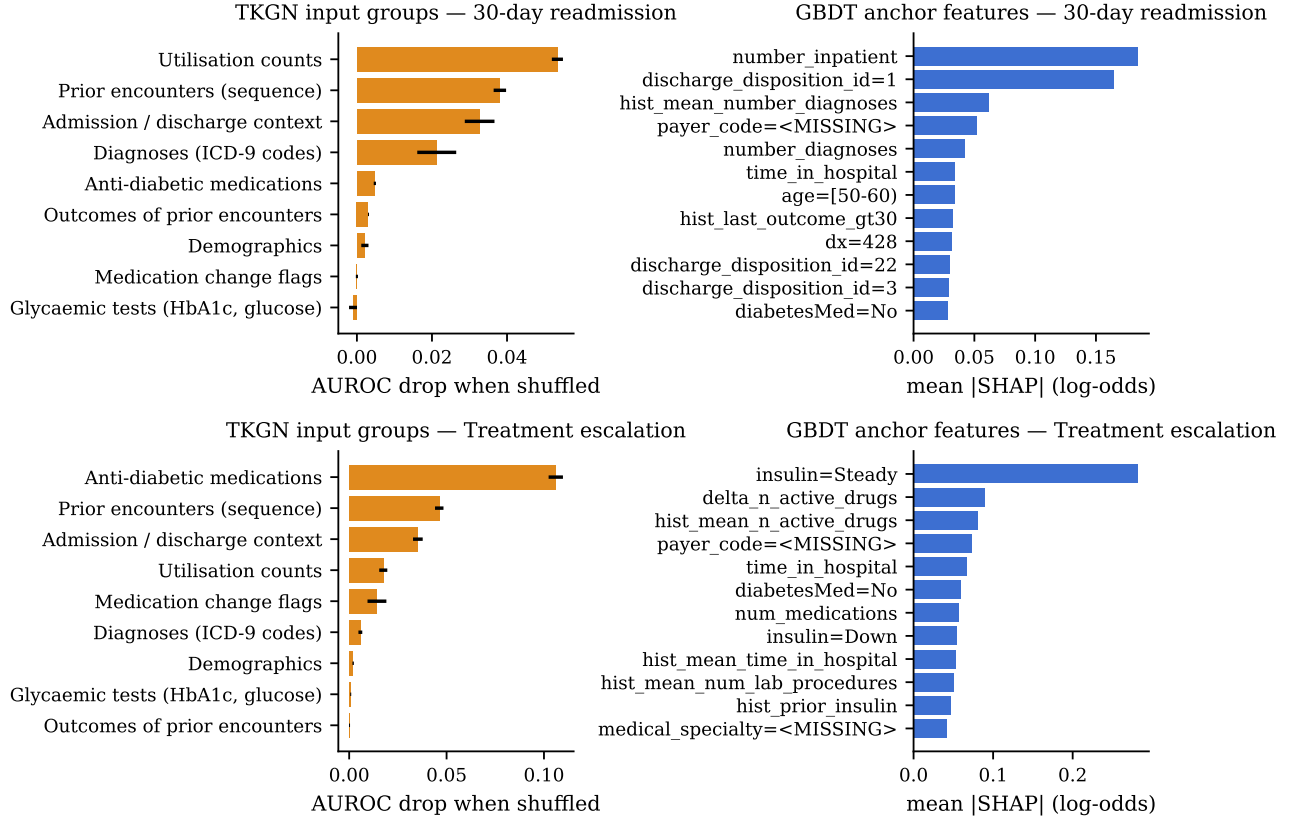


Figure 8: Left: permutation importance of input groups for TKGn (mean AUROC drop over three permutations). Right: mean absolute TreeSHAP value of the LightGBM anchor of TKGn-B. First grouped test set.

7.2. Interpretation

Why anchoring helps. The tabular learners already receive hand-crafted history aggregates, so the remaining signal that a sequence model can add is limited. Training the network on the residual of an out-of-fold boosting model concentrates its capacity on this remaining signal: the order and content of earlier stays, changes between stays, and patients whose history is long enough to saturate simple counts (Table 7). The same residual network trained on its own (TKGn) could not match boosting, which agrees with the broader observation that neural networks struggle to beat trees on heterogeneous tabular data (Grinsztajn et al., 2022). Combining both through an additive logit, rather than choosing between them, is a simple way to keep the strengths of each; the out-of-fold construction is essential, because in-sample boosting logits on the training set are optimistically sharp and would leave the network nothing to learn.

The role of clinical knowledge. Ontology-based code representations were originally motivated by rare codes and small datasets (Choi et al., 2017). In this cohort, most diagnosis codes are stored at the three-character level, and the frequent codes that carry most of the signal have hundreds to thousands of training examples, which leaves little room for hierarchical smoothing. The empirical co-morbidity propagation even reduced performance slightly, possibly because PPMI emphasises strongly associated but rare pairs (for example pregnancy-related categories) that are not predictive of the outcomes studied here. The dedicated data-efficiency experiment (Section 6.5) quantifies whether the graph helps when training data are scarce. We think

Table 7: Test AUROC by subgroup (first grouped repeat).

Subgroup	30-day readmission			Treatment escalation		
	<i>n</i>	TKGN-B	LightGBM	<i>n</i>	TKGN-B	LightGBM
<i>Race</i>						
AfricanAmerican	3,707	0.715	0.713	1,256	0.695	0.695
Asian	105	0.802	0.768	–	–	–
Caucasian	14,783	0.684	0.684	4,260	0.669	0.669
Hispanic	424	0.693	0.687	–	–	–
Other	316	0.736	0.727	–	–	–
Unknown	459	0.718	0.733	–	–	–
<i>Gender</i>						
Female	10,560	0.703	0.704	3,230	0.679	0.678
Male	9,234	0.680	0.678	2,539	0.679	0.679
<i>Age (years)</i>						
50-69	7,830	0.693	0.694	2,293	0.684	0.683
70+	8,736	0.660	0.659	2,595	0.682	0.682
<50	3,228	0.758	0.756	881	0.660	0.659
<i>Earlier stays</i>						
0	13,998	0.673	0.671	3,268	0.667	0.667
1	3,271	0.665	0.662	1,188	0.696	0.695
2+	2,525	0.620	0.608	1,313	0.693	0.692
<i>HbA1c</i>						
measured	3,387	0.675	0.681	914	0.705	0.704
not measured	16,407	0.695	0.693	4,855	0.674	0.674

the hierarchy is more likely to matter in settings with full-length ICD codes, smaller institutions or transfer between sites, and we report the negative result to discourage over-claiming of knowledge-graph benefits on this benchmark.

History-reliability gates. The gates did not change discrimination in the ablation, but they made the model’s use of history inspectable. For escalation they opened steadily with the length of the history, which matches the intuition that a longer medication trajectory is more informative for the next treatment decision. For readmission they damped the history summary while the explicit outcome tokens of earlier stays remained important, suggesting that “what happened after the last discharge” is more useful than a compressed summary of earlier stays.

Temporal robustness. The temporal split revealed shifts in recording practice (more diagnoses per stay) and in the population (longer histories, more emergency visits) that are typical of real hospital data. Such shifts are invisible to random splits and strongly penalised models that extrapolate linearly. They also explain why model rankings differed between designs. Any deployment of these models would need local recalibration and monitoring of feature distributions over time (Van Calster et al., 2019).

Fairness. Discrimination was similar for women and men and for the two largest racial groups, and calibration was acceptable in all adequately sized subgroups. The larger gap across age bands, with the weakest performance in the oldest patients, mirrors the higher and more homogeneous baseline risk of this group and deserves attention because older adults are a key target population for transitional-care programmes. Because the race variable is coarse and partly missing, these results do not rule out inequities for smaller groups (Obermeyer et al., 2019).

7.3. Comparison with previous work

The discrimination reached for 30-day readmission (AUROC around 0.69) lies at the upper end of the range typically reported for readmission models (Kansagara et al., 2011). Direct

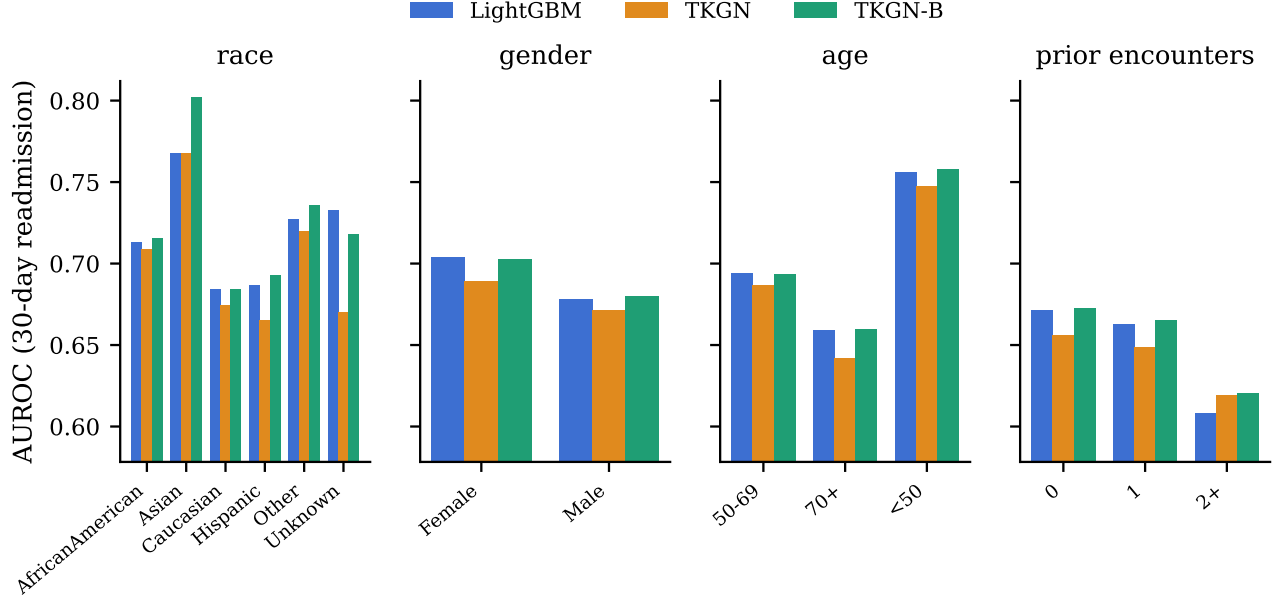


Figure 9: 30-day readmission AUROC by race, sex, age band and number of earlier stays.

comparison with earlier analyses of the same database is difficult because many use encounter-level random splits, which allow repeat stays of a patient to appear in training and test data, or restrict the cohort to one encounter per patient. We deliberately used patient-disjoint and temporal validation, multiple repeats and validation-only tuning, which give more conservative but more trustworthy estimates. Treatment escalation at the next hospital stay has, to our knowledge, not been used as a prediction target on this database; it links the readmission literature with work on clinical inertia in type 2 diabetes (Khunti et al., 2013).

7.4. Limitations

This study has several limitations. First, only one public data source was available; although the temporal split mimics prospective use, external validation in an independent health system is needed. Second, the release has no calendar dates, so stays were ordered by encounter identifier and the time between stays is unknown; time-aware models could exploit this information where it exists. Third, only three diagnoses per stay are recorded and most codes are truncated to three characters, which limits the potential of ontology-based representations. Fourth, readmissions are only observed within the participating systems, so some outcomes may be missed. Fifth, treatment escalation is defined from prescription states at the next inpatient stay and is therefore a proxy for glycaemic deterioration rather than a direct measure of it; it is only defined for patients who return to hospital. Sixth, the learning curves are based on a single repeat per training size, and the differences between the best models are small relative to their variability; claims of superiority over gradient boosting should therefore be made with caution. Seventh, the neural architectures were developed on a separate partition of the same cohort, so a small optimistic bias from design choices cannot be excluded; the tabular baselines were tuned only on the validation partitions of the evaluation runs. Finally, the models are research prototypes and have not been evaluated for clinical utility.

445 7.5. Conclusions

446 Using real, publicly available diabetes inpatient records and a leakage-controlled protocol,
447 we showed that anchoring a knowledge-gated sequence model on out-of-fold gradient boosting
448 yields a model that is consistently at least as good as the strongest tabular learners, clearly
449 better than standard deep sequence models, and well calibrated, for both 30-day readmission
450 and next-stay treatment escalation. Most of the gain from sequence modelling comes from the
451 patient’s earlier stays, whereas knowledge-graph code sharing did not improve discrimination at
452 full data size. All code, derived results and an interactive dashboard are released to support
453 replication and extension to other cohorts.

454 Data and code availability

455 The Diabetes 130-US Hospitals database is publicly available from the UCI Machine Learning
456 Repository (dataset 296, CC BY 4.0). All code needed to reproduce the cohort, experiments,
457 tables and figures, together with the derived result files, is available in the project repository.
458 The repository also contains an interactive dashboard and a REST service for exploring the
459 results; these are research prototypes and not medical devices.

460 Declaration of competing interest

461 The authors declare no competing interests.

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464 CRediT authorship contribution statement

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